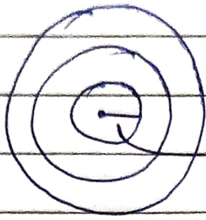
Date 09/01/2025

BSE

Biology → Basics → Gyan ki Baatein! (Lec No-1)

ORIGIN OF LIFE (Lec No-2)

Date 13/01/2025



→ Barysphere (core) (800-1400 miles)

Theories :-

1. Special Creation
2. Spontaneous Theory / ageonesis
3. Steady state Theory
4. Catastrophism
5. Panaspermia Theory / Cosmozoic Theory → {Spores}
6. Chemical Theory
   ↓
   oparin and Haldane

Nifo fekay hame

Modern Theory. (1953)
↳ Miller 18 days Exp.
↳ Chromatography / Calorimeter Technique
↓ sample containing
amino acid + Sugar + formaldehyde
↓ Str. formed
{Basic molecules produced} ← {Coaservates}

Date 16/01/2025 (Lec No-3)

Cell is a Basic :- Composed of Single cell
↳ Structural
↳ functional
↳ physiological

(× nucleus) ↑ {Nucleoid} {Prokaryotes} ← ↳ unicellular (Prokaryotic cell)

{Eukaryotes} ← and multicellular (Eukaryotic cell)

Eu → Developed { Pro → Primitive; Karyon → nucleus }

(controlling str. of a cell)

Bacteria present in the gut →

(Nerve cell) primitive cell, nucleoid or underdeveloped nucleus — **Living** — developed cell, well developed nucleus

Prokaryotes ← Arachrea → **Eukaryotes**

(# Arachrea showing quality of both prokaryotes and eukaryotes but more of prokaryotes)

Prokaryotes
- Bacteria
- Smaller → 1 to 5 μm in size
- (suspended nucleus or nucleoid containing hereditary material which contains DNA.) circular DNA
- No histone is present

Eukaryotes
- Bacteria present in: Plant, Animal, Fungi, Algae
- larger in size → 5 to 10 μm
- (spirally coiled, double standard, well developed) DNA.
- DNA + Histone protein (Base) — combined/complexed with

(Size of cell depends on the viscosity of the components and surface tension of the surface)

Unicellular — composed of single cell and all the activities are carried out in a single cell.

Cell Theory
(→ Small Rooms; containing of same cell type)

- 1665 → Robert Hook
- 1676 → Telescope discovery. they identified the cell. (Microbes) → ○

Plant cell are formed by a str.
- 1838 → Schleiden
- 1839 → Schwann — Animal cell (same observation)

Cell Theory or Schleiden & Schwann Theory
1. Cell is a structural, physiological organisation of living organism.
2. Cell can exist in dual form.
3. Cell is formed as a crystal → Go for crystallisation. Cell originates as a crystal form and with " next cell is formed, which is non-spontaneous in nature.

Modern Cell Theory :
↳ Rudolf Virchow
- cell is the basic unit of life
- All the cells have same chemical composition in an organism.
- cell originates from pre-existing cell with the help of cell division.

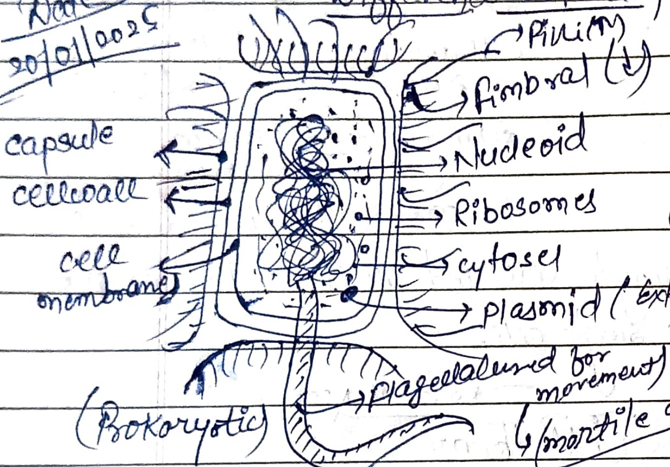
Difference between Plant cell and Animal cells

Plant cell	Animal cell
* cell wall is present — Large-Rigid structure and non-mobility.	* Cell covered with cell membrane.
* Chlorophyll present	* Chlorophyll absent
* Plastades present	* Plastades absent
* large vaccules are present.	* Vaccules are absent
	* Opposite of this.

(Nucleus are centrally positioned but due to this vaccuoles they are shifted from the centred)

Date 20/01/2025

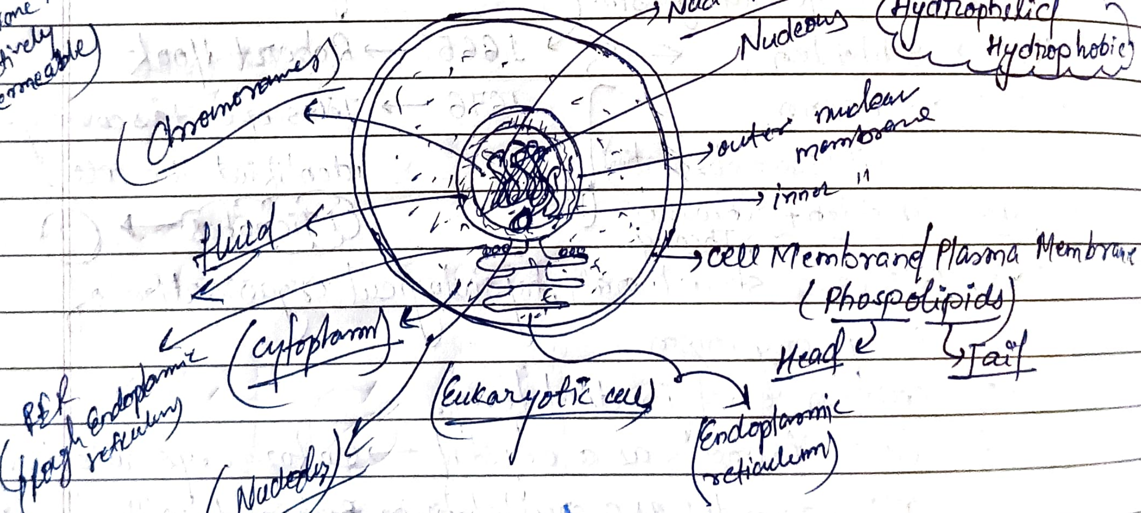
Difference between Prokaryotic and Eukaryotic cells (ter-on)



(Sex Pili)

1. Plasmamembrane/ cell membrane
2. cytoplasm
3. Cellular organelles.

(membrane is selectively permeable)



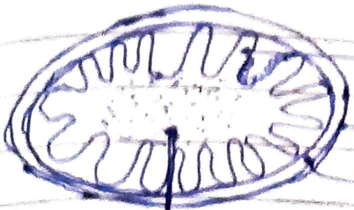
(Hydrophilic Hydrophobic)

70s 80s
Pro

Mitochondria:-

- Powerhouse of the cell.
- Provides energy for various purposes.

(Lec-05)



- Cristae (helps in ATP synthesis)
- OMM (outer mitochondrial membrane)
- IMM (Inner " ")
- Matrix

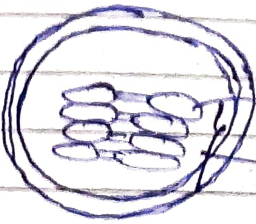
→ ETS chain

F_1-F_0 Particles

(#) glucose → ATP (Glycolytic cycle or Kreb's ")

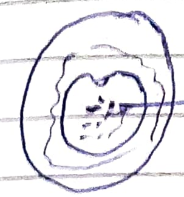
Chloroplast:-

(Responsible for Photosynthesis)



→ Thylakoid

→ Branches of ≡ is called granna (collection)



→ cytoplasm

Biomolecules

- Macromolecules
  - Protein
  - Carbohydrates
  - Lipids
  - Nucleic acids
- Micromolecules

(#) Nucleic acids → Biomol. are present in nucleus of a cell.
↳ Storage & transmission of genetic information.

- DNA
- RNA

(1) Nucleotide → Polynucleotide chain

* Sugar → ribose / Deoxyribose

DNA
↓ composed of
Nitrogen Base, Sugar, Phosphate group

Nitrogen Base → Purenes, Pyrimidines

Purenes → Adenine (A), Guanine (G) — (2) ring, Aromatic compound

Pyrimidines → (1) ring → Cytosine (C), Thymine (T)

DNA → ATGC

RNA → AUGC

Ribosome
↳ Protein factory

$\boxed{Nb} \rightarrow \boxed{S}$ → Nucleoside + P → tide

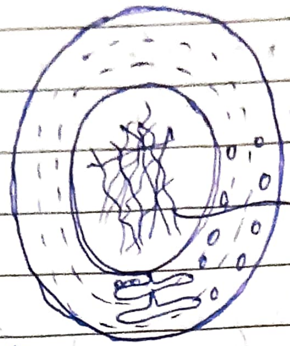
Date 27/01/2025

Wed → 05/02/2024 Bio Quiz

Center dogma

→ Heart of molecular Biology.

(Lec-06)



$$DNA \xrightarrow{\text{Transcription}} mRNA \xrightarrow{\text{Translation}} Protein$$

(always occurs in 3 to 5 dir.) ← Transcription + Translation = Centre Dogma

1. Initiation
2. Elongation
3. Termination

(Polymerization → of RNA → attachment of single unit again and again)

Synthesis of RNA → in the process DNA intercepts the message

(Splicing Editing Caping)

\# 3 nitrogen Base pairs make a code (Codon).

\# Humans have 64 codons.

AUGCGAUCAGC

AUG → (universal start codon)

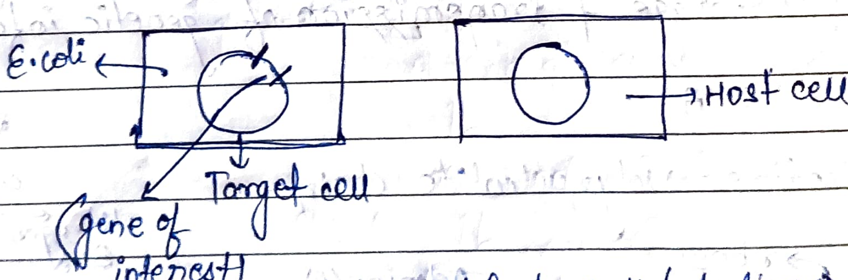
UAG, UAA, UGA → (universal stop codon.)

x ——— x ——— x ——— x ——— x ——— x

(Lec-07)

Date 30/01/2025

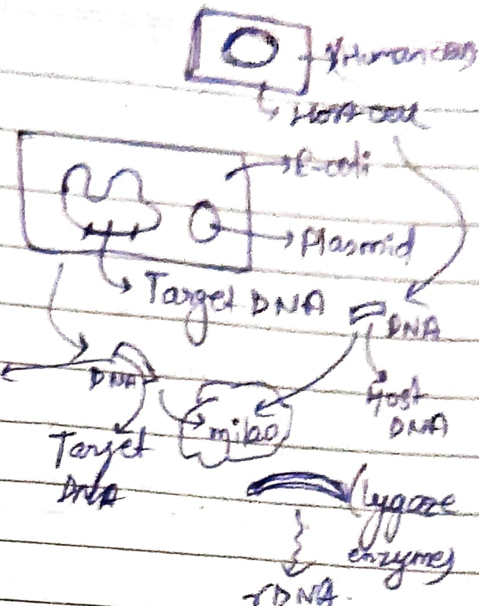
rDNA → Recombinant DNA Technology



1970 Boyer & cohan → first scientist to discover rDNA.

Paul Berg → Father of rDNA technology

How we will extract rDNA?
↳ ① Identification of Target DNA.
② Isolation of Target DNA.
③ Genomic DNA Extraction.
④ Restriction enzyme digestion.
⑤ ligation
⑥ Vector (Plasmid/cosmid/YAC)
↳ molecular vehicle to transfer one DNA from one place to another. In this plasmids are used.



⑦ Amplification of rDNA. (PCR)
⑧ Screening (Gel electrophoresis)
⑨ Gene expression

(1 lec-Bunk ↳ 07) (lec-08)

Cell Metabolism:-
life sustaining chemical transformation:-
① Conversion of food into energy
② " " " " nucleic acid/protein/lipid/carbohydrates.
③ " " " " wasteproducts (nitrogenous waste from body)

(Excretion)

Anabolic (Require some energy for large molecule synthesis.)

Catabolic → Release some energy (large molecule breakdown → small molecule synthesis.)

Aerobic
→ presence of O_2.
+nce of O_2
(obligate aerobic)

Anaerobic
→ absence of O_2
-nce of O_2.
(obligate ~~facultative~~ anaerobic).

(facultative anaerobic bacterial

Brokendown of food into energy (glucose) is called cell respiration.

Cell Respiration :-
(Glycolysis)

Describe glycolytic pathway / EMP Pathway

anything can be asked

EMP → Embadden Maye hoff parnas.

Glucose ⟶ Pyruvate

$C_6H_{12}O_6 \longrightarrow CO_2 + H_2O +$ Pyruvate

Glycolysis cycle Process

enzyme

Hexokinase ⟹ Glucose
⇓ ↳ ATP → ADP

Glucose-6-phosphate

Phospho hexokisminase ⟶ ⇓ ↳ ATP → ADP

Fructose-6-phosphate

Phosphofructo kinase ⟶ ⇓

fructose-1,6-bisphosphate

Triosephosphate dehydroxymatell ⇓ aldolase

Glyceraldehyde-3-phosphate | Dihydroxy acetone Phosphate

glyceraldehyde phosphate dehydrogenase ⟶ ⇓ ↳ NAD → NADH

1,2 bisphosphoglycerate

⇓

1,2 bisphosphoglycecate

Phospho glycenate kinase ⟶ ⇓ ↳ ADP → ATP

3-phosphoglycenate

Phospho glycenate mutase ⇓

Phosphoglycenate

⇓

Phosphoenolpyrovate

enalase ⇓ ↳ ADP → ATP ②

$\boxed{\text{Pyruvate}}$

1NADH → 3ATP
2NADH → 6ATP

Total ↳ 10 ATP